

Q. No.	Questions
UNIT 1	SECTION 1
1	A processor executes the following instructions: • MOV R1, #25 • MOV R2, 500 Initially, memory location 500 contains the value 80. Later, the value stored at memory location 500 is changed to 150, while the immediate operand remains unchanged. A processor designer wants to compare the behavior of Immediate and Direct Addressing Modes in terms of operand access and execution efficiency. Compare the execution of the two instructions by determining the values loaded into R1 and R2 before and after the memory modification. Recommend the addressing mode that provides greater flexibility and justify your recommendation. (5 Marks)
2	A processor executes branch and array-processing instructions using Relative and Indexed Addressing Modes. During execution, the Program Counter (PC) contains 1000, the Offset value is 50, the Address field is 500, and the Index Register contains 20. Later, the Index Register is updated to 40 while the Address field remains unchanged. Compare the effective addresses generated by Relative and Indexed Addressing Modes using the given information. Determine the effect of updating the Index Register from 20 to 40, and justify why Indexed Addressing is preferred for array-processing applications. (5 Marks)
3	A design team is developing a high-frequency event counter for a real-time signal processing unit. The current implementation "misplaces" a general-purpose ADD R1, #1 instruction inside a high-speed loop. The lead engineer proposes a redesign to use a specialized INC R1 instruction instead. Apply the Efficiency Principle to this scenario and analyze the impact on processor performance if the current arithmetic instruction requires three clock cycles to complete, whereas the proposed instruction requires only one. Justify your recommendation for the final hardware design based on the High-Performance Principle. (5 Marks)
4	(i) Consider the following instructions: • MOV R1,(R2)+ • MOV R3,-(R4) Assume • R2 = 500 and M[500] = 60 • R4 = 501 and M[500] = 80 Identify the values loaded into R1 and R3 and determine the final contents of R2 and R4 after execution. (3 Marks) (ii) A programmer needs to access consecutive memory locations during data transfer. Predict the addressing mode that reduce the number of explicit register update instructions and justify your answer. (2 Marks)
UNIT 1	SECTION 2

5 A company is developing a smart surveillance system that continuously processes video frames stored in consecutive memory locations. The processor must efficiently access each frame while minimizing instruction overhead.

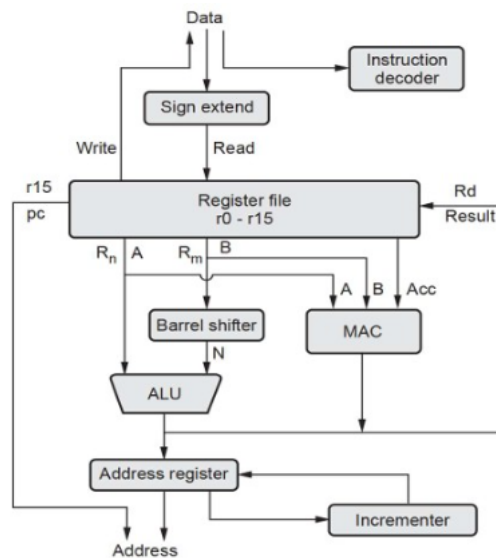
a) Identify the most suitable addressing mode for this application and discuss the generated effective address.

b) Justify the selection of the addressing mode with respect to execution efficiency.

(5 Marks)

6 A company is developing an intelligent industrial monitoring system based on an ARM processor. The system continuously collects data from sensors, performs arithmetic and logical operations, processes control signals, and stores intermediate results in registers. To achieve efficient processing with low power consumption, the designers have selected an ARM-based RISC architecture.

Sketch the basic ARM architecture for the given scenario and label its components.



(5 Marks)

7 A multimedia application repeatedly performs arithmetic operations on data already available in processor registers. A student suggests fetching operands from memory before every arithmetic operation instead of using the register contents. Refer to Figure 1.6.1 (32-bit ARM Architecture) to analyze the data flow between memory and registers.

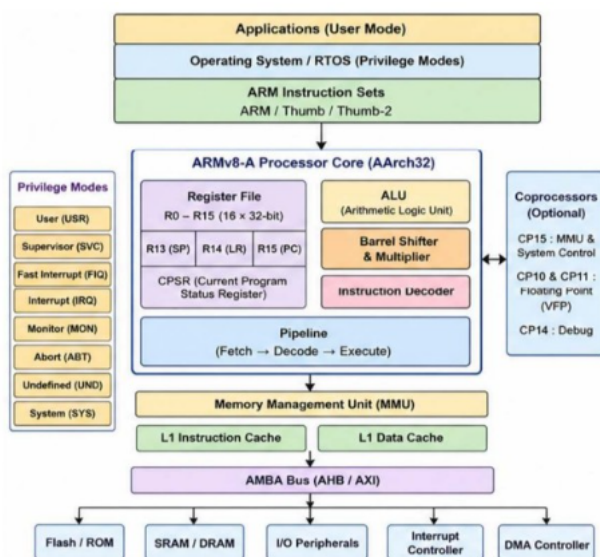
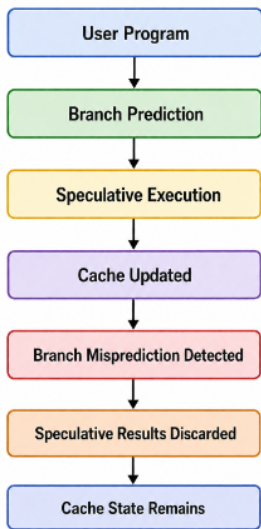


Figure 1.6.1: 32-bit ARM Architecture

Examine the effect of performing arithmetic operations using registers instead of accessing memory for every operation. Referring to Figure 1.6.1, justify a suitable strategy to reduce memory accesses and

	improve processor throughput. (5 Marks)
8	(i) A data analytics application processes very large memory spaces and performs intensive computations. The designer must decide whether to use x86 or x64 architecture. Identify the architecture best suited to support large address spaces and justify your answer. (3 Marks) (ii) A system designer wants additional registers without changing the existing x86 registers. Suggest a method for architectural enhancement and justify your answer. (2 Marks)
UNIT 2	SECTION 1
9	During the verification of a 32-bit single-cycle processor, an engineer monitors the control signals and hardware components activated while executing an unknown instruction. The following observations are recorded: 1. The Instruction Memory is accessed to fetch the instruction. 2. The Register File reads two source registers. 3. The Sign Extension Unit converts a 16-bit immediate value into a 32-bit value. 4. The ALU uses one register operand and the sign-extended immediate value to calculate an effective memory address. 5. The Data Memory performs a write operation using the data read from the second source register. 6. The Register Write (RegWrite) control signal remains disabled, indicating that no value is written back to the Register File. Based on these observations, identify the most likely instruction executed by the processor. Reconstruct the complete datapath flow from instruction fetch to memory access and justify the role of each hardware component and control signal involved in the execution. (5 Marks)
10	A wearable healthcare monitoring device uses a five-stage DLX pipelined processor to retrieve patient data from memory efficiently. The processor executes Load (LW) instructions to access sensor values stored in memory. The processor pipeline consists of the following stages: The diagram illustrates the five-stage DLX pipeline with the following components and flows: IF (Instruction Fetch): Fetch instruction from Instruction Memory using PC. The PC points to the Instruction Memory. ID (Instruction Decode & Register Read): Decode instruction and read source registers. The Register File provides two registers (Rs, Rt) to the ALU. EX (Execute / Address Calculation): Perform ALU operation or compute effective address. The ALU takes two operands (Operand 1, Operand 2) and produces a result (Result). MEM (Memory Access): Access data memory for load or store. The ALU result is used as the address for the Data Memory. The Data Memory provides data (Data) to the Register File. WB (Write Back): Write result back to destination register. The Register File writes the data back to the destination register. The overall flow is: PC → Instruction Memory → Register File → ALU → Data Memory → Register File. Figure : Five Stage DLX Pipeline One of the instructions executed by the processor is: LW R5, 16(R2) where R2 contains the base address of the patient data and 16 is the offset. Analyze the execution of the instruction LW R5, 16(R2) using the five-stage DLX pipeline shown above. Explain the function of each pipeline stage (IF, ID, EX, MEM, and WB) and describe how the instruction progresses through the pipeline until completion. (5 Marks)
11	A processor executes the following instructions: I1 : LW R1,0(R2)

	I2 : ADD R3, R1, R4 I3 : SUB R5, R6, R7 Analyze the pipeline execution and identify any hazards that occur. Explain how forwarding or stalling can be used to resolve the hazards. (5 Marks)
12	(i) A startup designs a single-cycle embedded processor with the following components: PC, Instruction Memory, Register File (2 Read Ports, 1 Write Port), ALU, Data Memory , Two Multiplexers, Sign Extension Unit. The processor must execute: add R3,R1,R2 lw R4,16(R5) sw R6,20(R5) beq R1,R2,Label Analyze the utilization of the datapath components for each instruction and determine whether the given datapath can execute all four instructions without hardware modification. Justify your answer. (3 Marks) (ii) Based on your analysis, identify any hardware resources that are shared within the single clock cycle (such as the ALU, Register File, or Memory). State whether this resource sharing creates any execution conflicts or affects the correctness of instruction execution. Justify your answer. (2 Marks)
UNIT 2	SECTION 2
13	A scientific computing application executes the following instruction sequence: LW R1, 0(R2) ADD R3, R1, R4 MUL R5, R3, R6 SUB R7, R5, R8 Analyze the data dependencies present in the instruction sequence and Infer how they influence the execution of multiple instructions in a modern processor. (5 Marks)
14	An operating system is running two applications on the same processor. The processor executes the following sequence.  <pre> graph TD A[User Program] --> B[Branch Prediction] B --> C[Speculative Execution] C --> D[Cache Updated] D --> E[Branch Misprediction Detected] E --> F[Speculative Results Discarded] F --> G[Cache State Remains] </pre> Figure : Preprocessor Execution Sequence

	Analyze the above execution flow and determine how sensitive information can still be inferred even after the speculative results are discarded. (5 Marks)
15	Analyze the following sequence used in a Spectre attack. Branch Prediction ↓ Speculative Execution ↓ Cache Update ↓ Timing Measurement ↓ Secret Information Leakage Explain how each stage contributes to the attack and identify where mitigation techniques can interrupt the attack. (5 Marks)
16	(i) A company is designing a high-performance processor for AI and scientific computing applications. To execute more instructions in parallel, the processor designer plans to increase the number of execution units (such as ALUs and Load/Store units). Show how the additional execution units enable more instructions to execute simultaneously and improve processor performance. (3 Marks) (ii) In the above scenario, Infer one factor that may still limit the processor's performance even after adding more execution units. (2 Marks)
UNIT 3	SECTION 1
17	A cloud data center is designed to handle millions of simultaneous user requests. The system uses multiple cache levels, large RAM capacity, and high-capacity SSD storage to improve processor efficiency and reduce memory access delays. Assess the contribution of different memory hierarchy levels in improving processor performance, and justify how the hierarchical organization reduces memory access delays. (5 Marks)
18	A National Online Examination Platform conducts computer-based examinations for thousands of students simultaneously. During peak examination hours, the processors frequently wait for data from memory, resulting in delays in loading question papers and saving student responses. A newly recruited engineer proposed removing the Cache Memory and increasing the Main Memory (RAM) capacity to improve system performance. Before implementing the proposal, the software architecture committee requested a technical evaluation based on the Memory Hierarchy concept.

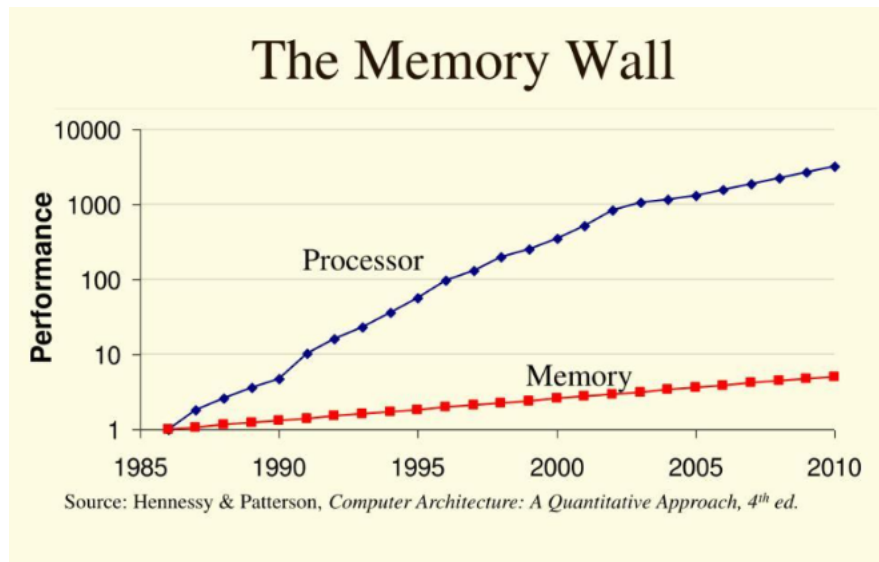


Figure: Processor-Memory Performance Gap

Analyze the Processor-Memory Performance Gap shown in the above figure and determine its impact on the performance of the online examination platform.

(5 Marks)

- 19** A hospital's Picture Archiving and Communication System (PACS) stores thousands of MRI scan images for diagnosis. To ensure rapid retrieval of patient images, the cache controller of the imaging server uses the Direct Mapping Technique.

During one retrieval cycle, the processor requests Memory Block 117. The cache memory contains 16 cache lines, and the controller must determine the exact cache line before checking whether the required data is available.

The system architect requests the engineering team to identify the cache location and explain the operation of direct mapping before deploying the server.

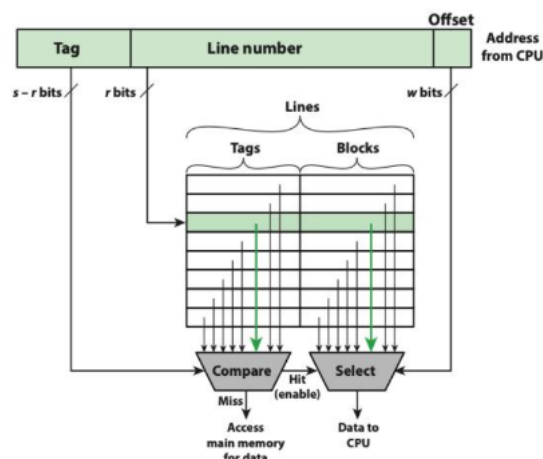


Figure: Direct Mapping Technique

Apply the Direct Mapping technique to determine the cache location for Memory Block 117. Justify your answer by explaining how the direct mapping mechanism shown in the reference figure enables the processor to retrieve the required data.

(5 Marks)

- 20** (i) A National Spacecraft Navigation and Mission Control Centre is developing an onboard computing system for a deep-space mission. The processor continuously performs orbital calculations, trajectory corrections, sensor fusion, and communication control. To improve execution speed, the engineering team reviews the modern processor cache hierarchy before approving the flight-qualified processor.

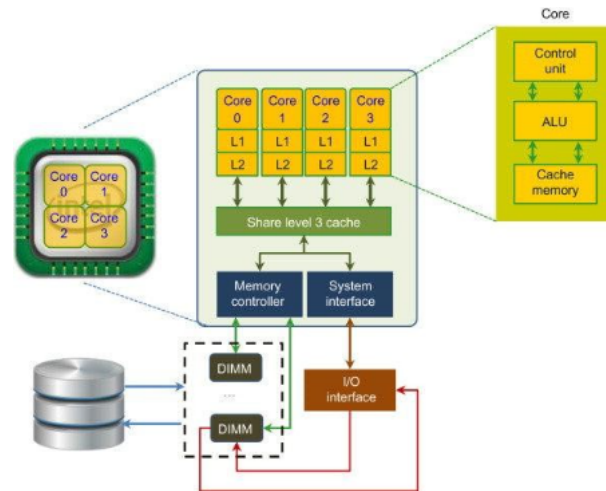


Figure: Modern Processor Cache Hierarchy

(a) Examine the Modern Processor Cache Hierarchy presented in the above figure.

(3 Marks)

(ii) (b) Assess the contribution of the Modern Processor Cache Hierarchy to continuous spacecraft navigation operations.

(2 Marks)